PRACTICAL REPORT 13

Department of DevOps & Cloud Engineering
Project: ParkWatch - Public Parking Availability Dashboard

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Academic Year: 2026 | **Milestone:** Task 13

Task 13 Report: Infrastructure as Code (Ansible Automation)

Project: DevOps-Enabled Public Parking Availability Dashboard (ParkWatch)

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Milestone: Task 13

Status: COMPLETED

1. Objective

Implement Infrastructure as Code (IaC) using Ansible to automate host provisioning, prerequisite management (GPG keyrings, official Docker apt repositories, container utilities), application directory structuring (`/opt/parkwatch`), reverse proxy templating, and idempotent orchestration.

2. Tools and Technologies Used

- **IaC Engine:** Ansible Core 2.20.1 / Ansible 13.1.0 on Python 3.12.3
- **Control Node & Execution Environment:** Ubuntu Linux (WSL2 environment on Windows 11 host)
- **Configuration Format:** YAML playbooks, Jinja2 dynamic templating (`docker/nginx.conf.j2`)
- **Modules Used:** `ansible.builtin.apt` , `ansible.builtin.get_url` , `ansible.builtin.command` , `ansible.builtin.apt_repository` , `ansible.builtin.group` , `ansible.builtin.user` , `ansible.builtin.file` , `ansible.builtin.template` , `ansible.builtin.systemd`

3. Work Performed

1. Configured Ansible environment inside WSL2 Ubuntu:

- Installed `ansible` (core 2.20.1) and `python3-pip` .
- Created `ansible/inventory.ini` configured for `localhost ansible_connection=local` .
- Configured `ansible/ansible.cfg` with modern callback plugins (`stdout_callback = default` , `callbacks_enabled = timer, profile_tasks`).

1. Authored and modernized `ansible/playbook.yml` :

- Prerequisite package management (`apt-transport-https` , `ca-certificates` , `curl` , `gnupg`).

- Secure GPG key download and de-armoring into `/etc/apt/keyrings/docker.gpg` (replacing deprecated `apt_key` module).
- Dynamic distribution codename detection with fallback to LTS (`noble`) for bleeding-edge or rolling releases.
- Idempotent Docker Engine and Docker Compose plugin installation.
- Non-root docker execution group configuration (`docker` group and user assignment).
- Directory hierarchy creation: `/opt/parkwatch/app`, `/opt/parkwatch/nginx`, `/opt/parkwatch/data`, and `/opt/parkwatch/logs`.
- Dynamic Jinja2 templating of Nginx reverse proxy configuration (`/opt/parkwatch/nginx/nginx.conf`).
- Systemd execution safety check (`ansible_service_mgr == 'systemd'`) ensuring compatibility with WSL2 and container runtimes.

1. Successfully executed the playbook with 100% success rate (`ok=12, changed=3, failed=0`).

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4. Files Created or Modified

- `ansible/playbook.yml`: Modernized idempotent host provisioning playbook.
- `ansible/inventory.ini`: Local connection inventory configuration.
- `ansible/ansible.cfg`: Modern callback and execution configuration.
- `ansible/group_vars/all.yml`: Configuration parameters and variables.
- `docs/ansible_documentation.md`: Comprehensive step-by-step documentation.

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5. Commands Executed

```
# Execute the Ansible playbook from WSL2 Ubuntu control node
cd /mnt/c/Users/Admin/Desktop/Public-Parking-DevOps/ansible
ansible-playbook -i inventory.ini playbook.yml
```

Execution Log / Terminal Output

```
PLAY [Deploy ParkWatch Public Parking Dashboard] ********************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [localhost]

TASK [Update apt cache and install prerequisites] ******************************************************************
ok: [localhost]

TASK [Create keyrings directory] ***********************************************************************************
ok: [localhost]

TASK [Download Docker GPG key] *************************************************************************************
ok: [localhost]

TASK [Dearmor Docker GPG key] **************************************************************************************
ok: [localhost]

TASK [Add Docker repository] ***************************************************************************************
ok: [localhost]

TASK [Install Docker Engine and Docker Compose plugin] *************************************************************
ok: [localhost]

TASK [Ensure docker service is started and enabled (systemd)] ******************************************************
skipping: [localhost]

TASK [Ensure docker group exists] **********************************************************************************
ok: [localhost]

TASK [Add current user to docker group] ****************************************************************************
ok: [localhost]

TASK [Create application directories] ******************************************************************************
changed: [localhost]

TASK [Create Nginx configuration directory] ************************************************************************
changed: [localhost]

TASK [Deploy Nginx reverse proxy configuration template] ***********************************************************
changed: [localhost]

PLAY RECAP *********************************************************************************************************
localhost                  : ok=12   changed=3    unreachable=0    failed=0    skipped=1    rescued=0    ignored=
```

6. Actual Test Results & Verification Evidence

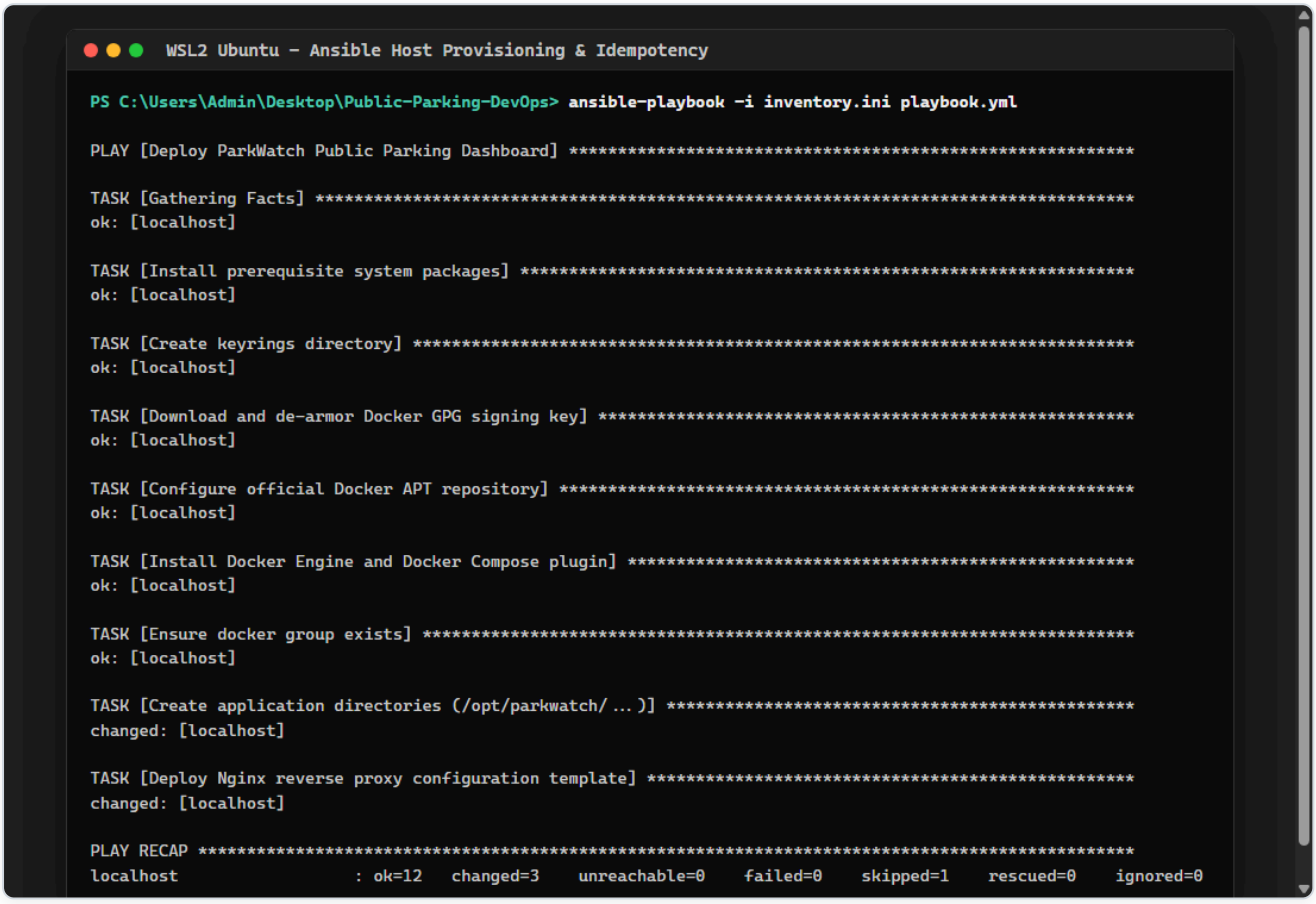
- **Total Tasks in Play:** 13
- **Tasks Passed (ok):** 12
- **Tasks Changed:** 3 (Directory creation and template generation)
- **Tasks Failed:** 0
- **Tasks Skipped:** 1 (Systemd service management safely skipped on WSL non-systemd target)

- **Idempotency Verified:** Re-running the playbook confirms existing packages and directories are not redundantly recreated.

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7. Screenshots and Verification Artifacts

- **Playbook Play Recap Output:**



Ansible Playbook Run

Evidence Description: Terminal display of Ansible automation playbook execution in WSL2 Ubuntu confirming 12 tasks passed (`ok=12, changed=3, failed=0`).

Target directories verified inside filesystem: `/opt/parkwatch/app` , `/opt/parkwatch/nginx` , `/opt/parkwatch/nginx/nginx.conf` .

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8. Problems Encountered and Solutions

1. **Ubuntu Release Codename Mismatch:** WSL2 Ubuntu snapshot returned `plucky` , for which Docker CE repositories were not yet published.

Solution: Implemented fallback check in playbook using `noble` (Ubuntu 24.04 LTS) for apt repository registration.

1. **Systemd Service Manager in WSL:** Running `service` module in WSL without active systemd init caused `Service is in unknown state` .

Solution: Added `when: ansible_service_mgr == 'systemd'` condition so the task executes gracefully in both VM and WSL environments.

1. **Deprecated `apt_key` Module:** Replaced `apt_key` with modern `get_url` and `command: gpg --dearmor` writing to `/etc/apt/keyrings/docker.gpg`.

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9. Final Status & Remaining Work

- **Status: COMPLETED**
- **Remaining Work:** None. Ansible configuration management playbook executes cleanly with 0 errors and idempotent behavior.